SmartRelay AI

Tagline: Predict failures before they happen.

One-line description: SmartRelay AI watches inverter relay behavior in real time and estimates when a relay is likely to fail, so teams can act before downtime.

Theme: Predictive maintenance for inverter relays using AI
Live demo: https://smartrelayapp.web.app/

2. Problem Statement

In many inverter installations, relay failures are discovered only after service drops. Teams can see raw readings, but they usually do not get a clear prediction of remaining relay life.

The result is reactive maintenance: urgent technician dispatch, replacement under pressure, and avoidable operational loss.

Pain points

- No practical “failure in X days” signal for service planning.
- Field teams are forced to respond after disruption begins.
- Hard to prioritize risk across multiple deployed inverters.

Real-world impact

- Downtime and SLA breach risk.
- Emergency replacement + technician cost.
- **Business loss example:** prevented failure saves **₹50,000** downtime cost.

3. Proposed Solution

SmartRelay AI combines live telemetry with a lightweight prediction model to estimate relay Remaining Life (%) and Failure Risk (%). It then converts this into simple actions: warning, critical, and service recommendation.

The design is split by purpose: edge logic for immediate local alerts, and cloud logic for prediction quality, trends, and fleet visibility.

Prediction objective

- **Inputs:** temperature, load %, switching cycles
- **Outputs:** Remaining Life (%), Failure Risk (%), Confidence Score, and “failure predicted in X days”

4. Key Innovations

Explainable AI

Shows **why** risk increases with a visible contribution split (temperature / load / cycles).

Real-time realism

Live charts plus **Real Data Mode** (mock API stream) with a “Connected to mock API” badge to remove demo ambiguity.

Edge + Cloud hybrid

Edge thresholds for immediate protection; cloud AI for remaining-life estimation and fleet analytics.

Story-driven replay

Historical failure replay slider to demonstrate progression from healthy → warning → high risk.

5. System Architecture

Implemented architecture flow:

Inverter → MODBUS → IoT Device → MQTT → Cloud → AI Model → Dashboard

Inverter + MODBUS

Registers expose relay temperature/load/current/voltage and switching behavior.

IoT Device + MQTT

Packages telemetry and streams it reliably to cloud ingestion.

Cloud + AI Model

Computes failure risk and remaining-life estimates continuously.

Dashboard

Turns predictions into alerts and technician actions for operations teams.

6. Model & Technical Approach

The current system uses a linear-regression-style risk scorer. We chose this deliberately: it is easy to explain, fast to run, and stable for real-time updates.

Inputs are temperature, load %, and switching cycles. Outputs are Remaining Life (%), Failure Risk (%), and Confidence Score. The same pipeline can be upgraded later to Random Forest or Gradient Boosting without redesigning the dashboard.

Inputs / outputs

- **Inputs:** Temperature, Load %, Switching cycles
- **Outputs:** Remaining Life (%), Failure Risk (%), Confidence Score
- **Explainability:** contribution breakdown for “Why did it fail?”

Real-world feasibility note

Tested with simulated inverter data aligned to MODBUS protocol ranges.

7. Features

Real-time dashboard

- Live charts updated every 1–2 seconds
- Prediction cards (risk, RUL, failure horizon)

Alerts + actions

- Warning / critical alerts
- **Field Engineer Mode:** `Send technician` with location + action

Multi-inverter intelligence

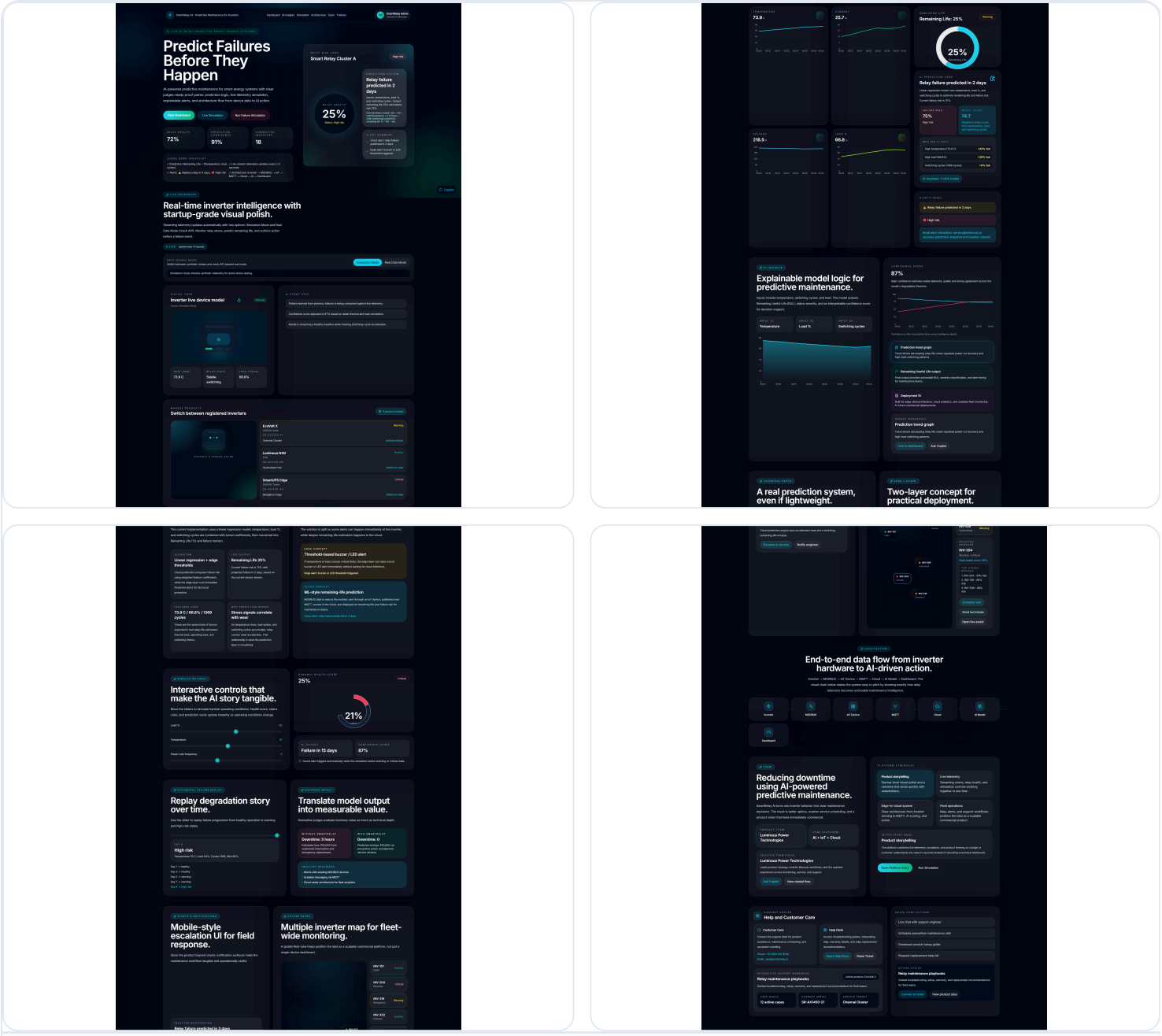
- Fleet health score
- Top 3 risky devices

Replay + insights

- Historical failure replay slider
- AI Assistant (1-click insight)
- Confidence vs Risk visualization

Evidence: Webpage Screenshots (Reference)

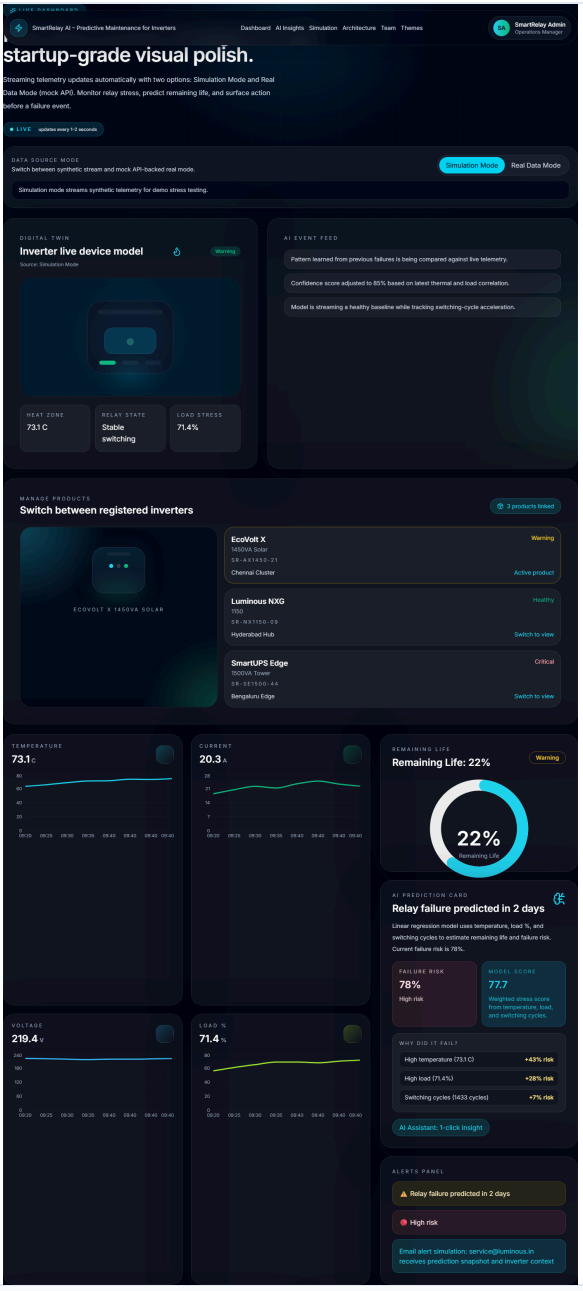
Screenshots below are from the current live deployment and map directly to the claims in this document.



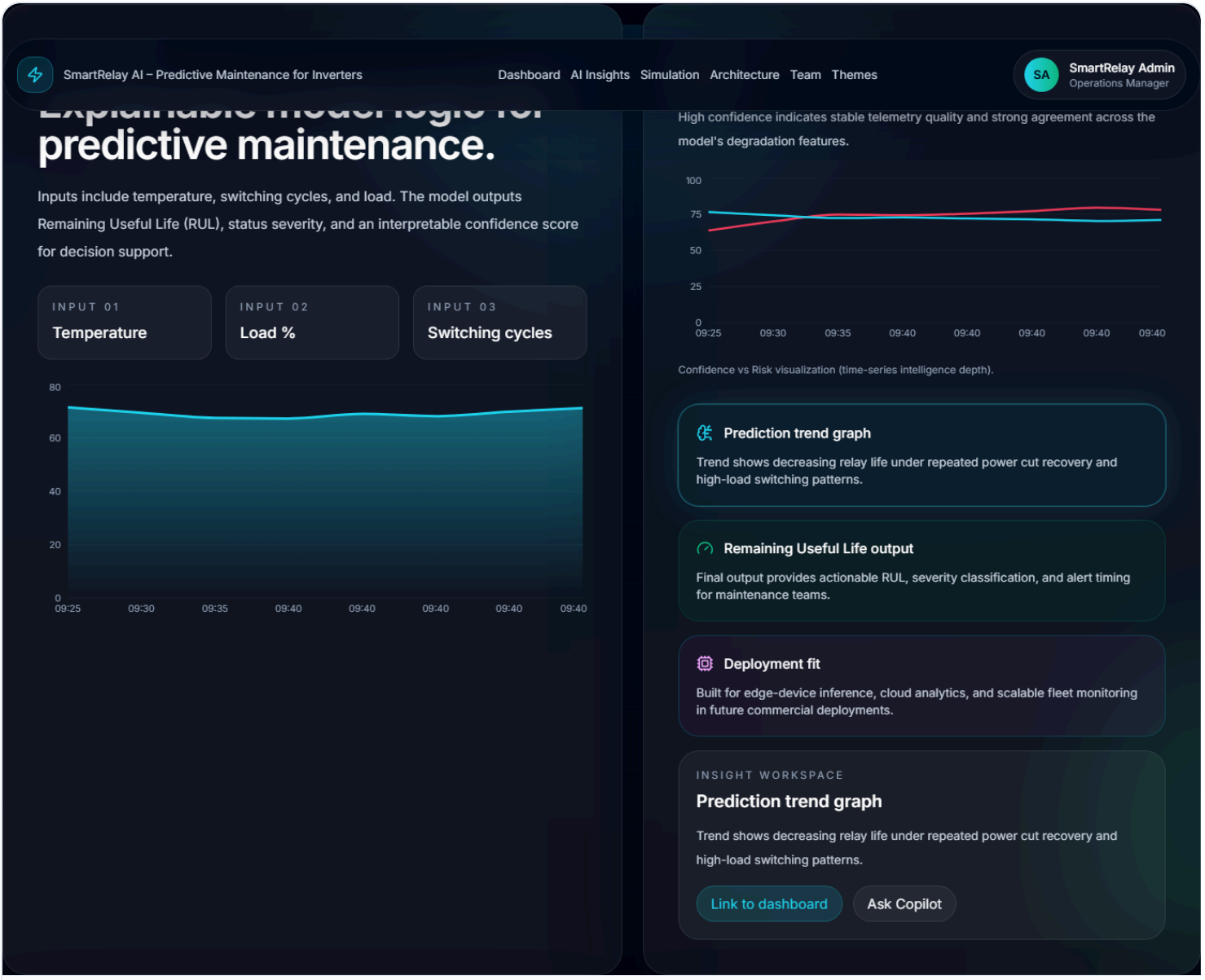
Dashboard overview split into four laptop-style captures: prediction cards, alerts, and live monitoring.

Additional Screenshot References

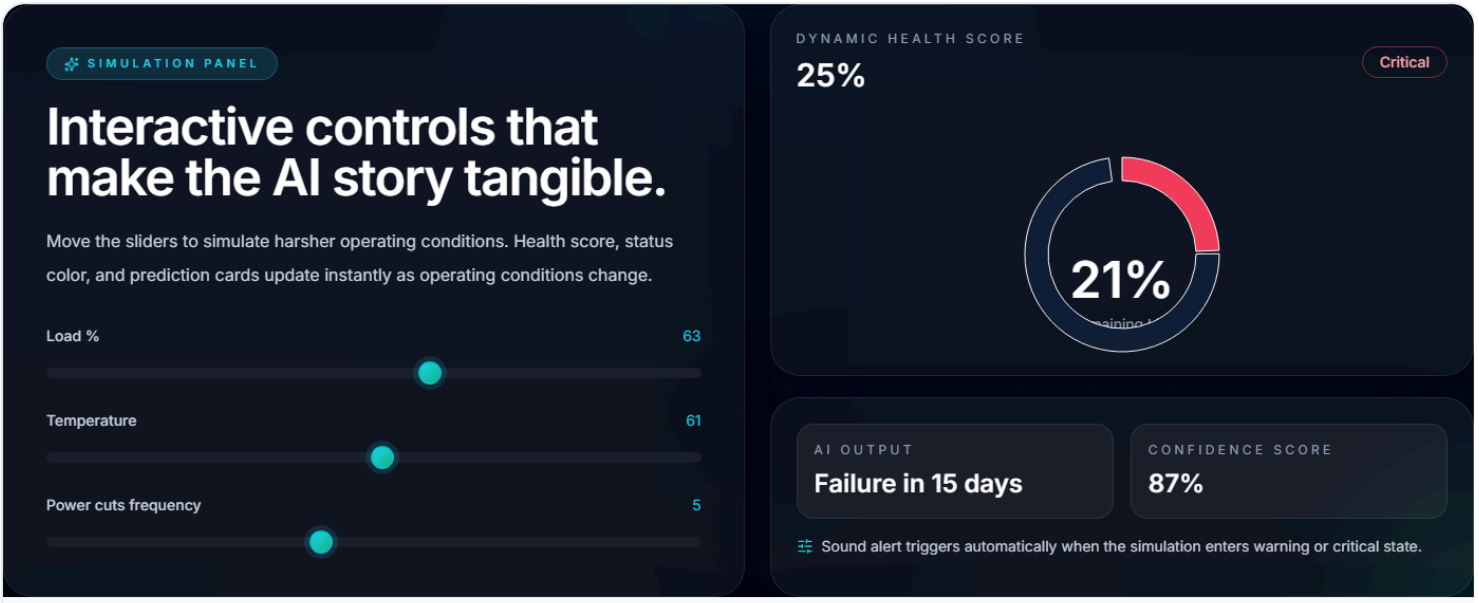
Supporting captures from specific sections of the live product.



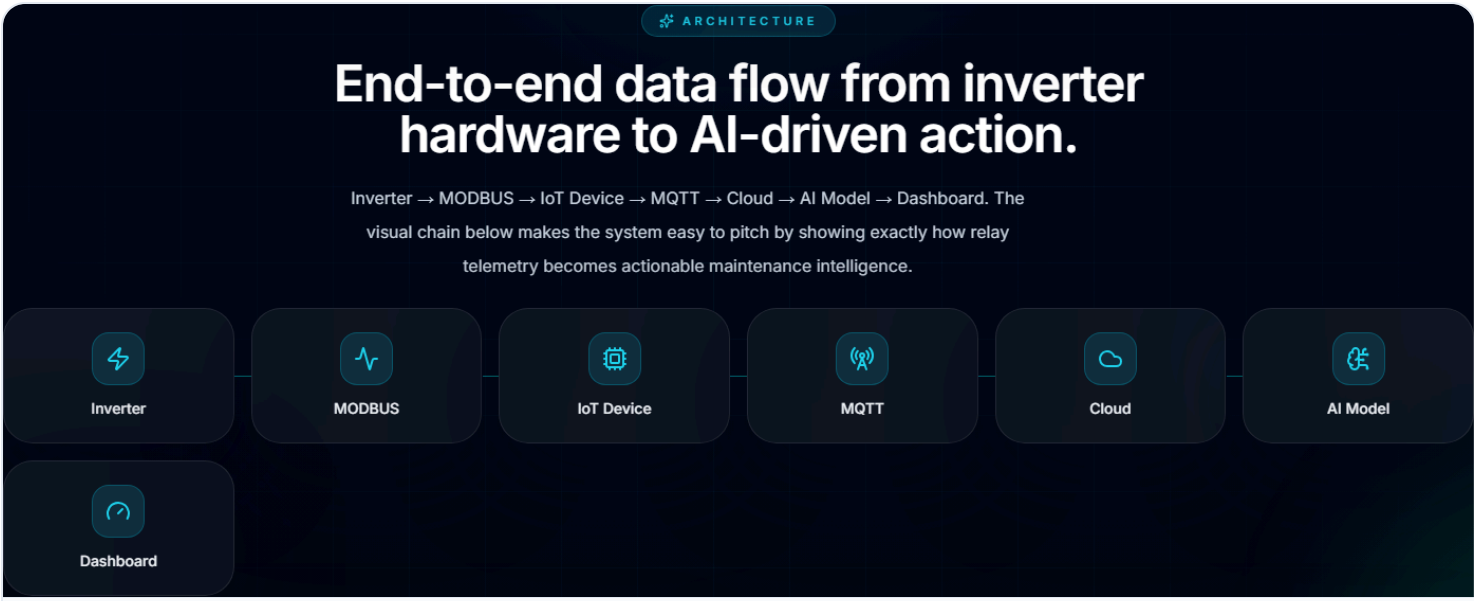
Live dashboard: streaming telemetry + data source mode toggle (Simulation / Real Data Mode) + mock API connectivity badge.



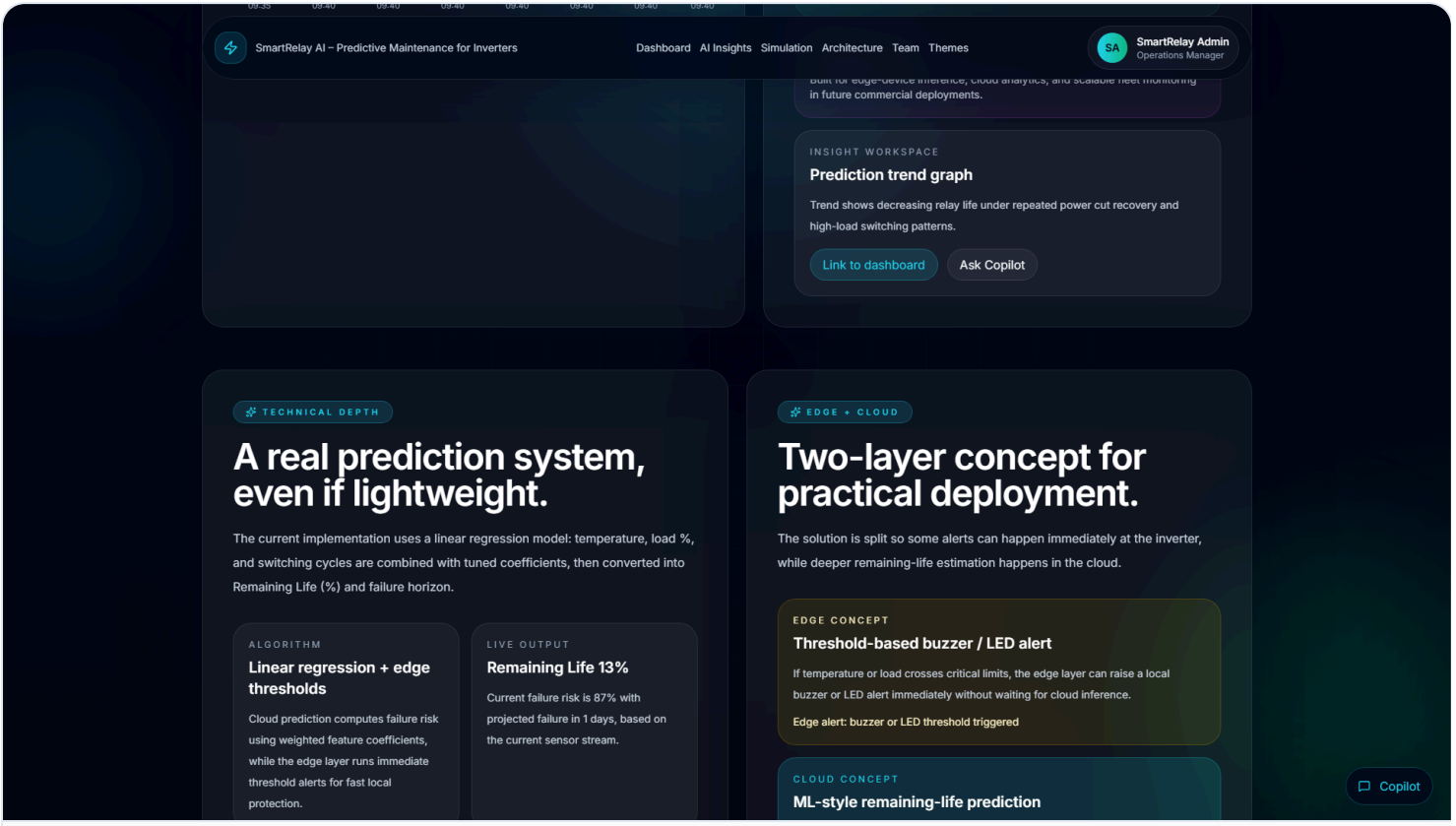
AI Insights: confidence and combined confidence-vs-risk trend visualization.



Simulation controls: interactive stress testing and health state changes.



Architecture: Inverter → MODBUS → IoT Device → MQTT → Cloud → AI Model → Dashboard.



Technical depth and Edge + Cloud concept for deployment feasibility.

10. Demo & Links

Live Website

https://smartrelayapp.web.app/

Submission readiness

- Live dashboard with real-time updates
- RUL + risk prediction with alerts
- Explainability and replay
- Fleet risk ranking and field action workflow